Isabel Esther Rodríguez Alvarez

Backend & AI Agent Engineer

Python · FastAPI · Microsoft Azure · RAG · Multi-Agent · MCP

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Summary

Software engineer with **5 years of experience** building **Python backends** and **production AI agents**. At La Nave, one of Cuba's most popular ride-hailing apps (**10,000+ users**), I build FastAPI services and event-driven Azure architectures and shipped **semantic and hybrid search** to production. As a freelancer, I design and build AI agents **from scratch** that run **24/7** for businesses in Panama, using RAG, multi-agent orchestration, MCP and function calling.

Experience

Backend & AI Developer · La Nave · Remote *Oct 2024 – Present*

Ride-hailing app connecting passengers with nearby drivers; 10,000+ users.

- Build and maintain the backend **REST APIs** in Python with **FastAPI**, with **JWT/OAuth2** authentication and **Redis** caching.
- Implemented app-to-backend communication with **Azure SignalR** (real time) and **Service Bus** (event-driven architecture) on Azure Functions and App Services.
- **Redesigned the address search** with **semantic and hybrid search** on Azure AI Search: it now suggests places and addresses in real time as users type, with more relevant, better-ranked results.
- **La Nave official store (in production):** built a Node.js/TypeScript plugin for semantic and hybrid product search and an **Azure AI Foundry agent** that automatically validates the product catalog.
- Design, model and manage **PostgreSQL** (Azure Database for PostgreSQL) and **Cosmos DB** (MongoDB API) databases.
- Automated deployments with **Docker**, Azure Container Registry and **Azure DevOps CI/CD** pipelines; monitor services with Application Insights.

AI Agent Developer · Freelance · Remote *2024 – Present*

Agents built from scratch for businesses in Panama: requirements with clients (English and Spanish), architecture, development, deployment and support.

- **ED Bienes Raíces (real estate): 24/7 WhatsApp and Telegram assistant** that guides clients, explains processes and **completes paperwork automatically**, replacing a fully manual workflow.
- **Restaurant chain:** supply-management agent that automates end to end **real-time inventory**, purchasing, sales and goods receiving.
- Architecture: **RAG** with vector and hybrid search, agent memory, **multi-agent orchestration**, function calling, **MCP** and streaming, with token-usage and LLM cost control.

Backend Developer · Datalisoft · Remote *Sep 2021 – Sep 2024*

Cuban software development company.

- Built backend APIs and services in Python with **FastAPI** and **Django**.
- Built **custom AI assistants** for clients across industries with **Azure OpenAI** and the Azure AI SDK, tailored to each business's data and processes.
- Developed **React** and **Streamlit** interfaces integrated with backend services.

Programming Instructor (social service) · UCI *Sep 2021 – Sep 2023*

- Taught OOP with Java, Data Structures I & II and Web Programming; labs with Spring and Django.

Technical Skills

Languages: Python, TypeScript, Java, C++
Backend: FastAPI, Django, Spring, Node.js, REST APIs, JWT/OAuth2, Redis, event-driven architecture
AI Agents & LLMs: Azure OpenAI (GPT), Claude, Llama, Mistral · LangChain, Microsoft Agent Framework, OpenAI Agents SDK, Azure AI Foundry · RAG, embeddings, vector and hybrid search, function calling, memory, multi-agent, MCP, streaming
Microsoft Azure: Functions, Service Bus, SignalR, App Services, AI Search, AI Foundry, Blob Storage, Azure Files, Container Registry
Databases: PostgreSQL, Cosmos DB / MongoDB, Redis
DevOps: Docker, Azure DevOps (CI/CD), Git, Application Insights
Frontend: React, TypeScript, Streamlit

Education

B.Sc. Computer Science Engineering · UCI, Cuba · **Gold Degree** (highest honors) *2021*

Languages

Spanish: native · **English:** B2, daily professional use (meetings, requirements and documentation with clients)